Inorganic and Materials Cumulative Exam
10/28/2022

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. (a) (14 pts) Two-dimensional (2D) materials (examples include graphene, MoS_2, MXene materials as we heard about recently in a Chemistry Seminar) are useful for their novel electronic properties. Describe the chemistry of bonding in these materials, including how they form bulk, three-dimensional crystals.

(b) (6 pts) The band gap of a 2D material (ex: MoS_2) is different in a bulk crystal than as a true, isolated 2D flake. Why is this?

2. Some 2D materials make excellent (electro)catalysts. The edge sites are often the catalytically active ones while the basal atoms are inert. Explain why this would be the case.

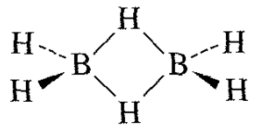
3. Describe an experiment using powder x-ray diffraction to measure the constant of thermal expansion of a crystalline solid. Please include any assumptions you may be making in your description.

4. (a) (12 pts) Describe the operating physical principle of a semiconductor-based photodiode used to measure light intensity. Include the details of the electronic structure of the semiconductor that enable the function.

(b) (8 pts) Which of the following semiconductors would be the best choice for a near-infrared photodiode (assume the wavelength of interest is 1000 nm)? - Al_2O_3 (band gap = 4.2 eV), InSb (0.17 eV), GaAs (1.7 eV), Ge (0.7 eV), TiO_2 (3.2 eV). To receive any credit for this question, you must explain your answer.

5. LiBr has the same crystal structure as NaCl. Li^+ has an ionic radius of 90 pm. Given that the density of LiBr is 3.464 g cm^{-3}, what is the ionic radius of Br^-?

6. Use the structure of diborane (provided) and your knowledge of symmetry to answer the following questions.



(a) (5 pts) What is the point group symmetry of diborane?

(b) (5 pts) Sketch the group orbitals for the bridging hydrogens in diborane.

(c) (5 pts) Based on the symmetry of diborane, you know the direction of the z-axis. Assume that the x-axis is in the plane of the page. Sketch the group orbitals made from the p_x orbitals on the boron atoms.

(d) (5 pts) Propose a molecular orbital diagram (for only these group orbitals – there are others) and sketch the appropriate combinations of the group orbitals you identified in (b) and (c).

7. (a) (5 pts) Consider the square pyramidal coordination complex $Ni(CN)_5^{3-}$. What is the point group of this complex?

(b) (8 pts) Modify the octahedral frontier orbital diagram to propose a molecular orbital diagram for the frontier orbitals of this square pyramidal complex.

(c) (7 pts) Based on your answer to (b) and the electronic structure of the complex, argue for why this complex would form a square pyramidal complex. You may use the attached periodic table to help with your answer.

8. (a) (10 pts) Describe the differences in electronic structure between TiCl4, an ionic solid, and TiO2, a covalent network solid.

(b) (10 pts) Given the orbital energies of two elements in a compound, how can you predict if they will form a covalent network solid or an ionic solid?

9. (a) (15 pts) Sketch the pressure-temperature, vapor-liquid-solid phase diagram of carbon dioxide. Label the phase regions in the diagram and any critical points.

(b) (5 pts) What is the name of the phase transition that occurs for solid CO_2 at 1 atm?

10. Use concepts of Lewis acid-base theory to explain why $MgCl_2$ is more soluble than $BaCl_2$ in water.

PERIODIC TABLE OF ELEMENTS

Atomic # / Symbol / Name / Weight

- C Solid
- Hg Liquid
- H Gas
- Rf Unknown

Metals: Alkali metals; Alkaline earth metals; Lanthanoids (Lanthanides); Actinoids (Actinides); Transition metals; Post-transition metals

Metalloids

Nonmetals: Other nonmetals; Noble gases

Pnictogens; Chalcogens; Halogens

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H Hydrogen 1.008																	2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122											5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305											13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

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